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S11-07

Final Journal

Assignment-1

Aim: Write a program for calculation of salary

Code:

```
import java.util.Scanner;
public class Salary
{
public static void main (String[] args)
{
int no,basic,da,hra,cca,pf,pt;
String name;
Scanner input=new Scanner(System.in);
System.out.println("Enter employee name, number and basic: ");
System.out.println();
name=input.nextLine();
no=input.nextInt();
basic=input.nextInt();
da=(int)(0.7*basic);
hra=(int)(0.3*basic);
cca=240;
pf=(int)(0.1*basic);
pt=100;
int grosssalary=basic+hra+da+cca-pf;
System.out.println("Gross salary is "+grosssalary);
}
}
```

Output:

Enter employee name, number and basic:

Aarya

40

600

Gross salary is 1380

Assignment-2

Aim: Write a program for implementing switch case.

Code:

```
import java.util.Scanner;

public class NumberOperations

{

 public static void main(String[] args)

{

 Scanner scanner = new Scanner(System.in);

 System.out.print("Enter a number: ");

 int number = scanner.nextInt();

 System.out.println("1. Check Prime");

 System.out.println("2. Check Even/Odd");

 System.out.println("3. Check Armstrong Number");

 System.out.println("4. Check Perfect Square");

 System.out.println("5. Check Palindrome");

 System.out.println("6. Calculate Factorial");

 System.out.println("7. Reverse Number");

 System.out.println("8. Calculate Sum of Digits");

 System.out.println("9. Print Triangular Pattern");

 System.out.println("10. Print Fibonacci Series");

 System.out.print("Enter your choice (1-10): ");

 int choice = scanner.nextInt();

 switch (choice)

{

 case 1:

 checkPrime(number);

 break;

 case 2:

 checkEvenOdd(number);

 break;
```

```java
case 3:
checkArmstrong(number);
break;
case 4:
checkPerfectSquare(number);
break;
case 5:
checkPalindrome(number);
break;
case 6:
System.out.println("Factorial of " + number + " is: " + factorial(number));
break;
case 7:
System.out.println("Reverse of " + number + " is: " + reverseNumber(number));
break;
case 8:
System.out.println("Sum of digits of " + number + " is: " + sumOfDigits(number));
break;
case 9:
printTriangularPattern(number);
break;
case 10:
printFibonacciSeries(number);
break;
default:
System.out.println("Invalid choice. Please choose a number between 1 to 10.");
break;
}
}
public static void checkPrime(int number)
{
boolean isPrime = true;
if (number <= 1)
isPrime = false;
```

```
else
{
for (int i = 2; i <= Math.sqrt(number); i++)
{
if (number % i == 0)
{
isPrime = false;
break;
}
}
}
System.out.println(number + " is" + (isPrime ? " a prime number." : " not a prime
number."));
}
public static void checkEvenOdd(int number)
{
System.out.println(number + " is" + (number % 2 == 0 ? " an even number." : " an odd
number."));
}
public static void checkArmstrong(int number)
{
int originalNumber = number;
int sum = 0;
int digits = (int) Math.log10(number) + 1;
while (number > 0)
{
int digit = number % 10;
sum += Math.pow(digit, digits);
number /= 10;
}
System.out.println(originalNumber + (sum == originalNumber ? " is an Armstrong
number." : " is not an Armstrong number."));
}
public static void checkPerfectSquare(int number)
```

```
{
 int sqrt = (int) Math.sqrt(number);
 System.out.println(number + (sqrt * sqrt == number ? " is a perfect square." : " is not a
perfect square."));
 }
 public static void checkPalindrome(int number)
{
 int originalNumber = number;
 int reverse = 0;
 while (number > 0)
{
 int digit = number % 10;
 reverse = reverse * 10 + digit;
 number /= 10;
 }
 System.out.println(originalNumber + (originalNumber == reverse ? " is a palindrome." : "
is not a palindrome."));
 }
 public static int factorial(int number)
{
 if (number == 0)
 return 1;
 else
 return number * factorial(number - 1);
 }
 public static int reverseNumber(int number)
{
 int reverse = 0;
 while (number > 0)
{
 int digit = number % 10;
 reverse = reverse * 10 + digit;
 number /= 10;
 }
```

```
return reverse;
}
public static int sumOfDigits(int number)
{
int sum = 0;
while (number > 0)
{
int digit = number % 10;
sum += digit;
number /= 10;
}
return sum;
}
public static void printTriangularPattern(int number)
{
for (int i = 1; i <= number; i++)
{
for (int j = 1; j <= i; j++)
{
System.out.print(j + " ");
}
System.out.println();
}
}
public static void printFibonacciSeries(int number)
{
int a = 0, b = 1;
System.out.print("Fibonacci Series: ");
for (int i = 0; i < number; i++)
{
System.out.print(a + " ");
int sum = a + b;
a = b;
b = sum;
```

```
 }
 System.out.println();
 }
}
```

Output:

Enter a number: 5

1. Check Prime

2. Check Even/Odd

3. Check Armstrong Number

4. Check Perfect Square

5. Check Palindrome

6. Calculate Factorial

7. Reverse Number

8. Calculate Sum of Digits

9. Print Triangular Pattern

10. Print Fibonacci Series

Enter your choice (1-10): 9

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5

Assignment-3

Aim: Write a program for implementing Method Overloading

Code:

```
public class MethodOverloading
{
   public static void main(String [] args)
  {
      System.out.println("The value of inches to centimeters is " +convertToCentimeters(10));
      System.out.println("The value of feet and inches to centimeter "+convertToCentimeters(12,18));
   }
   public static double convertToCentimeters(int xInches)
```

```
    {
       return 2.54*xInches;
     }
   public static double convertToCentimeters(int yFeet,int xInches)
          {
       double newInches= 12*yFeet;
       newInches +=xInches;
       return convertToCentimeters((int) newInches);
          }
}
```

Output:

The value of inches to centimeters is 25.4

The value of feet and inches to centimeter 411.48

Assignment-4

Aim: Write a program for implementing Constructor.

Code:

```
public class Constructors
{
 Constructors() //Default constructor
 {
 System.out.println("Default Constructor");
 }
 Constructors(int a, String name) // Parameterized constructor
 {
 System.out.println("The constructor with two parameters INTEGER and
STRING, the values are "+ a+ ", " +name);
 }
 Constructors(boolean x)
 {
 this(7, "Aarya");
 System.out.println("The statement is "+x);
 }
 public static void main(String []args)
```

```
{
 Constructors obj= new Constructors();
 Constructors obj1= new Constructors(8,"Aarya");
 Constructors obj2= new Constructors(true);
 }
}
```

Output:

Default Constructor

The constructor with two parameters INTEGER and STRING, the values are 7,

Aarya

The constructor with two parameters INTEGER and STRING, the values are 8,

Aarya

The statement is true

Assignment-5

Aim: To implement vectors and strings

Code:

```
public class ProgramString{
public static void main(String[] args){
String s1= "Aarya is my name";
String s2= "Im in java lab";
String s3= “AARYA is my name";
String s4= "Im learning java";
System.out.println(" The length of the string s1 :" +s1.length());
System.out.println(" The length of the string s2 :" +s2.length());
 System.out.println(" The length of the string s3 :" +s3.length());
System.out.println(" The length of the string s4 :" +s4.length());
System.out.println(" The index of the string s1 :" +s1.charAt(3));
System.out.println(" The index of the string s2 :" +s2.charAt(4));
System.out.println(" The index of the string s3 :" +s3.charAt(5));
System.out.println(" The index of the string s4 :" +s4.charAt(6));
System.out.println(" The uppercase of the string s1 :" +s1.toUpperCase());
System.out.println(" The lowercase of the string s1 :" +s1.toLowerCase());
System.out.println(" The uppercase of the string s2 :" +s2.toUpperCase());
```

```
System.out.println(" The lowercase of the string s2 :" +s2.toLowerCase());
System.out.println(" The uppercase of the string s3 :" +s3.toUpperCase());
 System.out.println(" The lowercase of the string s3 :" +s3.toLowerCase());
System.out.println(" The uppercase of the string s4 :" +s4.toUpperCase());
System.out.println(" The lowercase of the string s4 :" +s4.toLowerCase());
System.out.println(" The concatination of the string s1 :" +s1.concat(s2));
System.out.println(" The concatination of the string s2 :" +s2.concat(s3));
System.out.println(" The concatination of the string s3 :" +s3.concat(s4));
System.out.println(" The concatination of the string s4 :" +s4.concat(s1));
System.out.println(" The substring of the string s1 :" +s1.substring(9,11));
System.out.println(" The substring of the string s2 :" +s2.substring(6,10));
System.out.println(" The substring of the string s3 :" +s3.substring(0,5));
System.out.println(" The substring of the string s4 :" +s4.substring(3,11));
System.out.println(" The replacement of the string s1 :"
+s1.replace("my","his"));
System.out.println(" The replacement of the string s2 :"
+s2.replace("in","outside"));
System.out.println(" The replacement of the string s3 :" +s3.replace(
"SOHAM","soham"));
System.out.println(" The replacement of the string s4 :"
+s4.replace("learning","pursuing"));
System.out.println("The index of the string s1 "+s1.indexOf('n'));
System.out.println("The index of the string s1 "+s2.indexOf('j'));
System.out.println("The index of the string s1 "+s3.indexOf('M'));
System.out.println("The index of the string s1 "+s4.indexOf('a'));
String s5= "" , s6= "";
s5= s1.replace("Aarya","Soham");
s6=s1.concat("he is best");
System.out.println("The index of the string s1 "+s5);
System.out.println("The index of the string s1 "+s6);
}
}
```

Output:

The length of the string s1 :16

The length of the string s2 :14

The length of the string s3 :16

The length of the string s4 :16

The index of the string s1 :a

The index of the string s2 :n

The index of the string s3 :

The index of the string s4 :r

The uppercase of the string s1 :AARYA IS MY NAME

The lowercase of the string s1 :aarya is my name

The uppercase of the string s2 :IM IN JAVA LAB

The lowercase of the string s2 :im in java labT

he uppercase of the string s3 :AARYA IS MY NAME

The lowercase of the string s3 :aarya is my name

The uppercase of the string s4 :IM LEARNING JAVA

The lowercase of the string s4 :im learning java

Assignment-6

Aim: To implement inheritance

Code:

```
class Book{
public static void main(String[] args) {
Book book = new Book();
book.display("How to win friends and influence people", "Dale Carnegie",
1936, 1, 550.0);
Reference_Book ref = new Reference_Book();
ref.display("Applied Physics Part II", "I.A. Shaikh", 2000, 2, 329.0);
Magazine mag = new Magazine();
mag.display("Vogue", "Anonymous", 2004, 5, 250.5);
}
protected String bookName;
protected String authorName;
protected int publishedYear;
protected int editionNo;
```

```java
protected double cost;
public void display(String bookName, String authorName, int publishedYear,
int editionNo, double cost) {
System.out.println("The name of book is " + bookName);
System.out.println("The author of book is " + authorName);
System.out.println("The year of publishing is " + publishedYear);
System.out.println("The edition number is " + editionNo);
System.out.println("The cost of book is " + cost);
}
}
class Reference_Book extends Book{
public void display(String bookName, String authorName, int publishedYear,
int editionNo, double cost) {
System.out.println();
System.out.println("This is a reference book");
System.out.println("The name of book is " + bookName);
System.out.println("The author of book is " + authorName);
System.out.println("The year of publishing is " + publishedYear);
System.out.println("The edition number is " + editionNo);
System.out.println("The cost of book is " + cost);
}
}
class Magazine extends Book{
public void display(String bookName, String authorName, int publishedYear,
int editionNo, double cost) {
System.out.println();
System.out.println("This is a magazine");
System.out.println("The name of book is " + bookName);
System.out.println("The author of book is " + authorName);
System.out.println("The year of publishing is " + publishedYear);
System.out.println("The edition number is " + editionNo);
System.out.println("The cost of book is " + cost);
}
}
```

Output:

The name of book is How to win friends and influence people

The author of book is Dale Carnegie

The year of publishing is 1936

The edition number is 1

The cost of book is 550.0

This is a reference book

The name of book is Applied Physics Part II

The author of book is I.A. Shaikh

The year of publishing is 2000

The edition number is 2

The cost of book is 329.0

This is a magazine

The name of book is Vogue

The author of book is Anonymous

The year of publishing is 2004

Assignment-7

Aim: To implement interface

Code:

```
interface vehicle

{

void brake();

void tyre();

void move();

}

class bike implements vehicle

{

public void brake() { System.out.println("bike brakes are fine");}

public void tyre() { System.out.println("bike tyres are fine");}

public void move() { System.out.println("bike moves");}

}

class car implements vehicle

{
```

```
public void brake() { System.out.println("car brakes are fine");}
public void tyre() { System.out.println("car tyres are fine");}
public void move() { System.out.println("car moves");}
}
class bicycle implements vehicle
{
public void brake() { System.out.println("bicycle brakes are fine");}
public void tyre() { System.out.println("bicycle tyres are fine");}
public void move() { System.out.println("bicycle moves");}
}
class auto
{
public static void main(String args[])
{
bike a=new bike();
a.brake();
a.tyre();
a.move();
car b=new car();
b.brake();
b.tyre();
b.move();
bicycle c=new bicycle();
c.brake();
c.tyre();
c.move();
}
}
```

Output:

bike brakes are fine

bike tyres are fine

bike moves

car brakes are fine

car tyres are fine

car moves

bicycle brakes are fine

bicycle tyres are fine

bicycle moves

Assignment-8

Aim: To implement Packages

Code:

```
package letmecalculate;
public class Calculator {
public int add (int a, int b)
{
return a+b;
}
public int multiply (int a, int b)
{
return a*b;
}
public int subtract (int a , int b)
{
return a-b;
}
public int divide (int a , int b)
{
return a/b;
}
public int modulo (int a , int b)
{
return a%b;
}
}
package pattern;
public class PatternMaker {
```

```
public void patternOf(int rows)
{
int i,j,k,m;
for (i=1;i<=rows; i++)
{
for (j=1; j<=rows-i; j++)
{
System.out.print(" ");
}
for (k=1; k<=i; k++)
{
System.out.print("*");
}
for(m=1;m<i;m++){
System.out.print("*");
}
System.out.println();
}
for (i=1;i<=rows; i++)
{
for (j=1; j<=i; j++)
{
System.out.print(" ");
}
for (k=1; k<=rows-i; k++)
{
System.out.print("*");
}
for(m=1;m<rows-i;m++){
System.out.print("*");
}
System.out.println();
}
}
```

```
}
import letmecalculate.*;
import pattern.*;
import java.util.*;
public class Demo {
public static void main (String[] args)
{
Calculator random = new Calculator();
Scanner s = new Scanner(System.in);
int m,n;
System.out.println("Enter Number1 : ");
m = s.nextInt();
System.out.println("Enter Number2 : ");
n = s.nextInt();
int a = random.add(m,n);
int b = random.subtract(m,n);
int c = random.multiply(m,n);
int d = 0;
int e = 0;
boolean exc = false;
try { d = random.divide(m,n);
e = random.modulo(m,n);
}
catch (ArithmeticException ae)
{ System.out.println("Denominator cannot be 0");
exc = true;
}
System.out.println("Addition: " + a + "\nSubtraction: " + b + "\nMultiplication: " + c);
if (!exc) { System.out.println("\nDivision: " + d+ "\nModulo: " + e);
}
PatternMaker random = new PatternMaker();
System.out.println("Enter number of rows: ");
int x = s.nextInt();
random.patternOf(x);
```

```
}
```

Assignment-9

Aim: To implement Multithreading

Code:

```
class Hello implements Runnable{
public void run()
{
for(int i=0; i<5; i++)
{
System.out.println("hello");
try{
Thread.sleep(500);
}catch(Exception e){
};
}
}
}
class Hi implements Runnable{
public void run()
{
for(int i=0; i<5; i++)
{
System.out.println("hi");
try{
Thread.sleep(500);
}
catch(Exception e){
};
}
}
}
class MultithreadingDemo
{
```

```
public static void main(String []args)
{
Hi obj1 = new Hi();
Hello obj2 = new Hello();
Thread t1 = new Thread(obj1);
Thread t2 = new Thread(obj2);
t1.start();
t2.start();
System.out.println(t1.isAlive());
System.out.println(t2.isAlive());
try{
t1.join();
}
catch(Exception e){
};
try{
t2.join();
}catch(Exception e){
};
System.out.println(t1.isAlive());
System.out.println(t2.isAlive());
System.out.println("Bye");
}
}
```

Output:

```
hi
true
true
hello
hi
hello
hi
hello
```

hi

hello

hi

hello

false

false

Bye

Assignment-10

Aim: To implement exception handling

Code:

```
public class JavaExceptionExample {
public static void main(String args[]){
try{
//code that may raise exception
int data=100/0;
System.out.println("rest of the code...");
}
catch(ArithmeticException e) {
System.out.println(e);
}
}
}
```

Output:

java.lang.ArithmeticException: / by zero

Assignment-11

Aim: To implement I/O Streams

Code:

```
import java.io.*;
class Main
{
public static void main(String[] args) throws IOException
{
```

```
String text;
BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
System.out.println("Enter First number");
text = br.readLine();
int a = Integer.parseInt(text);
System.out.println("Enter second number");
text = br.readLine();
int b = Integer.parseInt(text);
int c = a + b;
System.out.println("The sum is : " + c);
}
}
```

Output:

Enter First number 5

Enter second number 3

The sum is : 8

Assignment-12

Aim: Write a program for creating a calculator.

Code:

```
import java.awt.*;
import java.awt.event.*;
class Calculator extends Frame implements ActionListener
{
 Frame f=new Frame();
 Label l=new Label("Enter number one");
 Label l1=new Label("Enter number two");
 Label l2=new Label("Result");
 Button b=new Button("Add");
 Button b1=new Button("Subtract");
 Button b2=new Button("Multiply");
 Button b3=new Button("Divide");
 Button b4=new Button("Cancel");
 TextField t=new TextField();
```

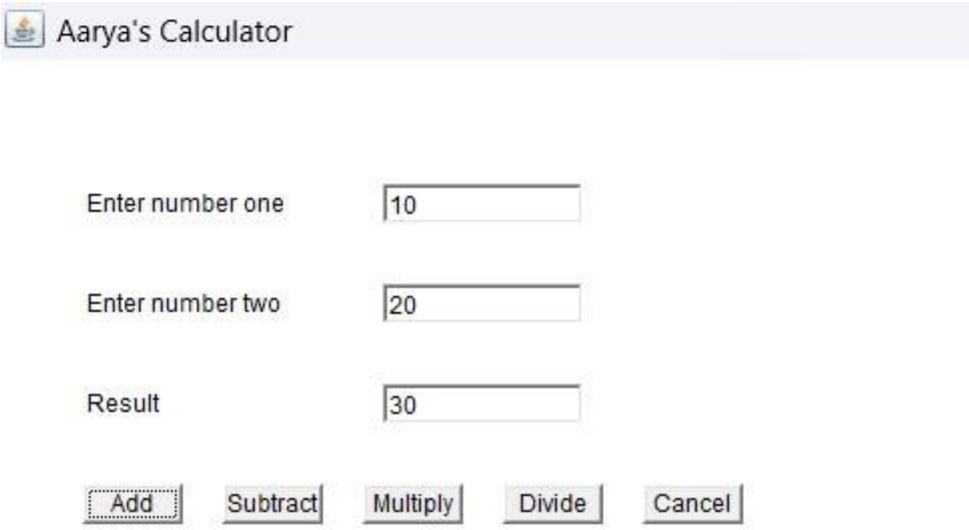
```
TextField t1=new TextField();
TextField t2=new TextField();
Calculator()
{
l.setBounds(50,100,100,20);
l1.setBounds(50,150,100,20);
l2.setBounds(50,200,100,20);
b.setBounds(50,250,50,20);
b1.setBounds(120,250,50,20);
b2.setBounds(190,250,50,20);
b3.setBounds(260,250,50,20);
b4.setBounds(330,250,50,20);

t.setBounds(200,100,100,20);
t1.setBounds(200,150,100,20);
t2.setBounds(200,200,100,20);
f.add(b);
f.add(b1);
f.add(b2);
f.add(b3);
f.add(b4);
f.add(t);
f.add(t1);
f.add(t2);
f.add(l);
f.add(l1);
f.add(l2);
b.addActionListener(this);
b1.addActionListener(this);
b2.addActionListener(this);
b3.addActionListener(this);
b4.addActionListener(this);
f.setTitle("Aarya's Calculator");
f.setLayout(null);
```

```
f.setSize(500,500);
f.setVisible(true);
}
public void actionPerformed(ActionEvent e)
{
int n1=Integer.parseInt(t.getText());
int n2=Integer.parseInt(t1.getText());

if(e.getSource()==b)
{
t2.setText(String.valueOf(n1+n2));
}
if(e.getSource()==b1)
{
t2.setText(String.valueOf(n1-n2));
}
if(e.getSource()==b2)
{
t2.setText(String.valueOf(n1*n2));
}
if(e.getSource()==b3)
{
t2.setText(String.valueOf(n1/n2));
}
if(e.getSource()==b4)
{
System.exit(0);
}
}
public static void main (String[] args)
{
Frame f=new Calculator();
}
}
```

Output:



Assignment-13

Aim: Write a program for creating a Student form

Code:

```
import javax.swing.*;
import java.awt.event.*;
public class Studentform extends JFrame implements ActionListener
{
JLabel l1,l2,l3,l4,l5;
JTextField t1,t2;
JRadioButton r1,r2;
JComboBox cb;
JCheckBox c1,c2,c3;
JButton b;
JTextArea area;
Studentform()
{
 JFrame f=new JFrame(" Student Form");
 JLabel l1=new JLabel("Name: ");
```

```
l1.setBounds(20,20,80,30);
t1=new JTextField();
t1.setBounds(100,20,150,30);
f.add(l1); f.add(t1);

JLabel l2=new JLabel("Mobile No.: ");
l2.setBounds(20,70,80,30);
t2=new JTextField();
t2.setBounds(100,70,150,30);
f.add(l2); f.add(t2);

JLabel l3=new JLabel("Gender: ");
l3.setBounds(20,120,80,30);
r1=new JRadioButton("Male");
r1.setBounds(100,120,60,30);
r2=new JRadioButton("Female");
r2.setBounds(180,120,120,30);
ButtonGroup bg=new ButtonGroup();
bg.add(r1); bg.add(r2);
f.add(r1); f.add(r2); f.add(l3);

JLabel l4=new JLabel("Age");
l4.setBounds(20,165,80,30);

String age[]={"18","19","20","21","22","23"};
cb=new JComboBox(age);
cb.setBounds(100,170,90,20);
f.add(l4); f.add(cb);

JLabel l5=new JLabel("Hobbies");
l5.setBounds(20,215,50,30);
f.add(l5);
c1=new JCheckBox("Reading");
c1.setBounds(80,220,80,20);
```

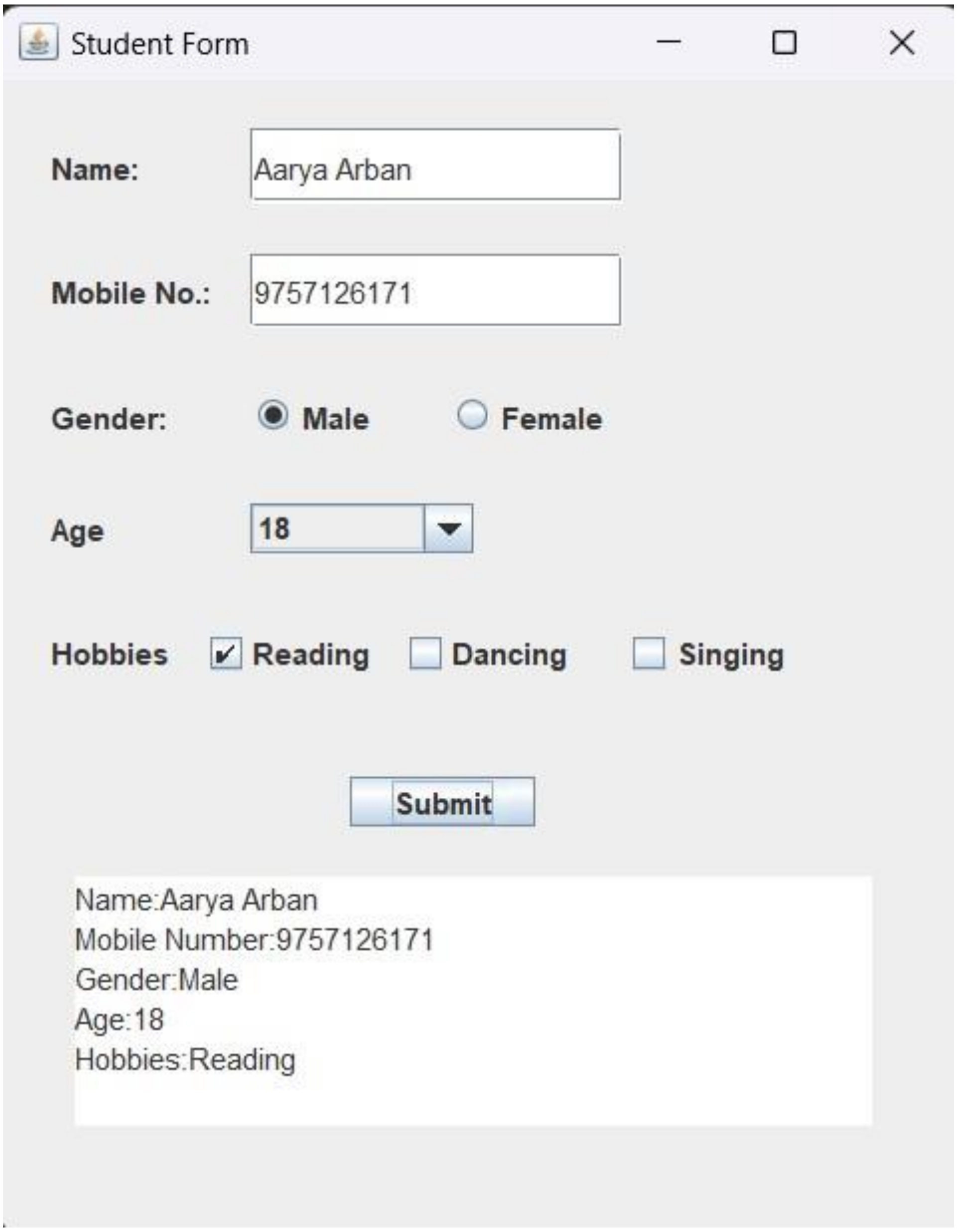
```java
 c2=new JCheckBox("Dancing");
 c2.setBounds(160,220,80,20);
 c3=new JCheckBox("Singing");
 c3.setBounds(250,220,80,20);
 f.add(c1); f.add(c2); f.add(c3);
 JButton b=new JButton("Submit");
 b.setBounds(140,280,75,20);
 f.add(b);

 area=new JTextArea();
 area.setBounds(30,320,320,100);
 f.add(area);
 b.addActionListener(this);

 setDefaultCloseOperation(EXIT_ON_CLOSE);
 f.setSize(400,500);
 f.setLayout(null);
 f.setVisible(true);
}
public void actionPerformed(ActionEvent e)
{
 String name=t1.getText();
 String mobile=t2.getText();
 String gender= r1.isSelected() ? "Male": r2.isSelected() ? "Female":"-";
 String age= cb.getItemAt(cb.getSelectedIndex()).toString();
 String hobby="";
 if(c1.isSelected())
 {
 hobby="Reading";
 }
 if(c2.isSelected())
 {
 hobby=hobby+" "+"Dancing";
 }
```

```
if(c3.isSelected())
{
hobby=hobby+" "+"Singing";
}
area.setText("Name:"+name+"\n"+"Mobile
Number:"+mobile+"\n"+"Gender:"+gender+"\n"+"Age:"+age+"\n"+"Hobbies:"+hobby);
}
public static void main(String[] args)
{
new Studentform();
}
}
```

Output:



Assignment-14

Aim: Write a program for creating a Menubar

Code-1:

```
import java.awt.*;
class Notepad
{
Notepad()
{
Frame f= new Frame("MenuBar");
MenuBar mb= new MenuBar();
f.setMenuBar(mb);
Menu fi= new Menu("File");
mb.add(fi);
MenuItem N= new MenuItem("New");
MenuItem O= new MenuItem("Open");
MenuItem S= new MenuItem("Save");
MenuItem SA= new MenuItem("SaveAs");
MenuItem E= new MenuItem("Exit");
fi.add(N);
fi.add(O);
fi.add(S);
fi.add(SA);
fi.add(E);
Menu ed= new Menu("Edit");
mb.add(ed);
MenuItem C= new MenuItem("Cut");
MenuItem Co= new MenuItem("Copy");
MenuItem P= new MenuItem("Paste");
MenuItem D= new MenuItem("Delete");
ed.add(C);
ed.add(Co);
ed.add(P);
```

```
ed.add(D);
Menu vi= new Menu("View");
mb.add(vi);
MenuItem Z= new MenuItem("Zoom");
Menu submenu= new Menu("Zoom");
MenuItem in= new MenuItem("Zoom in");
MenuItem out= new MenuItem("Zoom out");
submenu.add(in);
submenu.add(out);
vi.add(submenu);
 MenuItem Sb= new MenuItem("Statusbar");
vi.add(Z);
vi.add(Sb);
Menu hp= new Menu("Help");
mb.add(hp);
MenuItem vh= new MenuItem("View help");
MenuItem an= new MenuItem("About Notepad");
 MenuItem sf= new MenuItem("Send Feedback");
 hp.add(vh);
hp.add(an);
hp.add(sf);
f.setVisible(true);
f.setSize(400,400);
f.setLayout(null);
}
public static void main(String []args)
{
new Notepad();
}
}
```

Output:

Code-2:

```
import javax.swing.*;

import java.awt.event.*;

public class NewMenubar implements ActionListener

{

JFrame f;

JMenuBar mb;

JMenu file, edit, help;

JMenuItem cut, copy, paste, selectAll;

JTextArea ta;

NewMenubar()

{

f=new JFrame();

mb= new JMenuBar();

f.setJMenuBar(mb);

f.add(mb);

file= new JMenu("File");

edit= new JMenu("Edit");

help= new JMenu("Help");

mb.add(file);

mb.add(edit);

mb.add(help);

cut= new JMenuItem("Cut");

 copy= new JMenuItem("Copy");

paste= new JMenuItem("Paste");

selectAll= new JMenuItem("Select all");

edit.add(cut);
```

```
edit.add(copy);
edit.add(paste);
edit.add(selectAll);
cut.addActionListener(this);
copy.addActionListener(this);
paste.addActionListener(this);
selectAll.addActionListener(this);
ta= new JTextArea();
ta.setBounds(20,20 ,360, 320);
f.add(ta);
f.setLayout(null);
f.setSize(400,400);
f.setVisible(true);
}
public void actionPerformed(ActionEvent e)
{
if(e.getSource()==cut)
ta.cut();
if(e.getSource()==copy)
ta.copy();
if(e.getSource()==paste)
ta.paste();
if(e.getSource()==selectAll)
ta.selectAll();
}
public static void main(String[] args)
{
new NewMenubar();
}
}
```

Output:

Assignment-15

Aim: Student Form using JavaFX

Code:

```
import javafx.application.Application;
import javafx.event.ActionEvent;
import javafx.event.EventHandler;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.control.Label;
import javafx.scene.control.PasswordField;
import javafx.scene.control.TextField;
import javafx.scene.layout.VBox;
import javafx.stage.Stage;
public class FirstJavaFX extends Application {
 public void start(Stage primaryStage) throws Exception {
 VBox vb = new VBox();
 vb.setSpacing(10);
 vb.setAlignment(Pos.CENTER);
 Label l1 = new Label("Your Username");
 TextField tx1 = new TextField();
 tx1.setMaxWidth(160);
 Label l2 = new Label("Your Username");
 PasswordField tx2 = new PasswordField();
 tx2.setMaxWidth(160);
```

```
Button button = new Button("Login");
TextField tx3 = new TextField();
tx3.setMaxWidth(160);
vb.getChildren().addAll(l1, tx1, l2, tx2, button, tx3);
button.setOnAction(new EventHandler<ActionEvent>() {
@Override
public void handle(ActionEvent arg0) {
String userName = tx1.getText();
String password = tx2.getText();
if (userName.equals("TSEC") && password.equals("bandra")) {
tx3.setText("Login Successful");
} else {
tx3.setText("Invalid User");
}
}
});
Scene scene = new Scene(vb, 600, 400);
primaryStage.setTitle("First JavaFX Application");
primaryStage.setScene(scene);
primaryStage.show();
}
public static void main(String[] args) {
launch(args);
}
}
```

Assignment – 16:

Writeup – 1:

Ghar Ka Khana

Aim: To create order

management system

Write-Up:

This Java code defines a class called "Order" within the "system.model" package. The class represents an order in a restaurant management system.

An "Order" object has three attributes:

orderID: An integer that uniquely identifies the order. price: A

double representing the total price of the order. date: A string

indicating the date when the order was placed.

The constructor for the "Order" class takes these attributes as parameters and initializes an "Order" object with them.

In summary, this code encapsulates the essential information related to an order, such as its ID, price, and the date it was placed. It provides a blueprint for creating order objects within the restaurant management system, facilitating the tracking and management of customer orders.

This Java code represents a graphical user interface for a order management system. It's designed to help restaurant staff take and process customer orders efficiently.

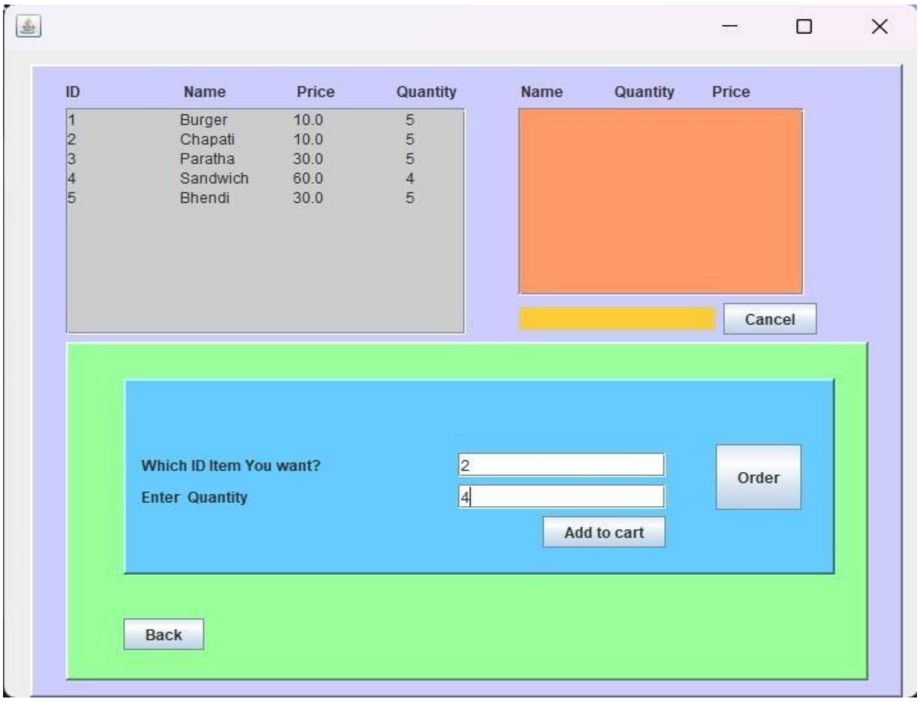
The code initializes the interface, connects to services for handling items and orders, and displays a list of available menu items to the user. Customers can add items to a virtual cart, and the code keeps track of the total price as items are added.

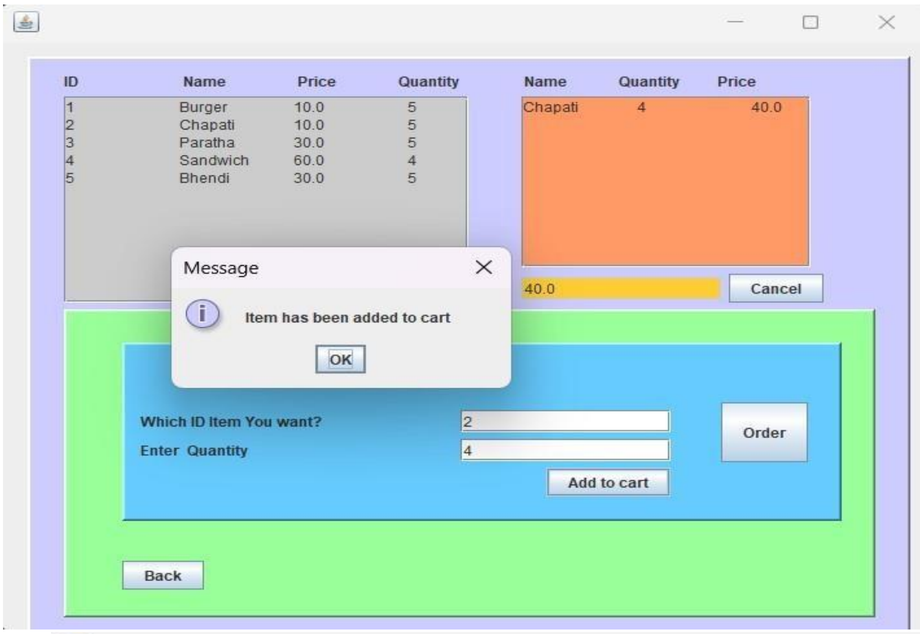
When the user decides to place an order, the code records the order details, including the items and their quantities, in files. It also updates the inventory to reflect the items that have been ordered.

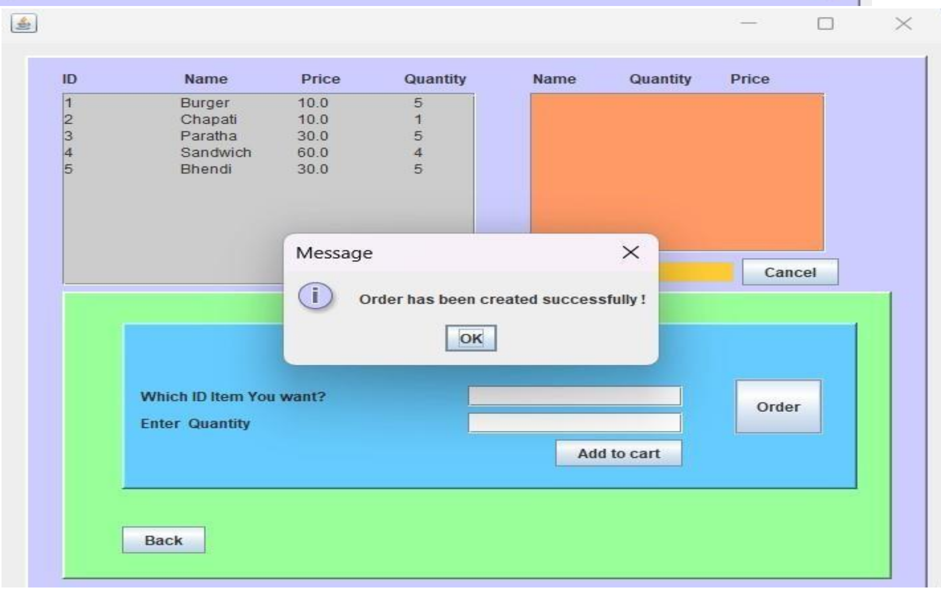
The code allows for backtracking to the main menu and clearing the cart as needed. It's a work-in-progress, as evidenced by some commented-out lines and opportunities for improvement.

In summary, this code is a key component of a order management system, streamlining the order-taking process and ensuring accurate records of customer orders and inventory management.

Output:


ID
Name
Price
Quantity
1 Burger 10.0 5
2 Chapati 10.0 5
3 Paratha 30.0 5
4 Sandwich 60.0 4
5 Bhendi 30.0 5
Name
Quantity
Price
Cancel
Which ID Item You want?
2
Enter Quantity
4
Add to cart
Order
Back


ID
Name
Price
Quantity
1 Burger 10.0 5
2 Chapati 10.0 5
3 Paratha 30.0 5
4 Sandwich 60.0 4
5 Bhendi 30.0 5
Name
Quantity
Price
Chapati 4 40.0
40.0
Cancel
Message
Item has been added to cart
OK
Which ID Item You want?
2
Enter Quantity
4
Add to cart
Order
Back


ID
Name
Price
Quantity
1 Burger 10.0 5
2 Chapati 10.0 1
3 Paratha 30.0 5
4 Sandwich 60.0 4
5 Bhendi 30.0 5
Name
Quantity
Price
Cancel
Message
Order has been created successfully !
OK
Which ID Item You want?
Enter Quantity
Add to cart
Order
Back

Code:

```
package system.model;
/**
 *
 * @author Aarya
 */ public class Order {
private int orderID;
private double price;
private String date;
   public Order(int orderID, double price, String date) {
this.orderID = orderID;
this.price = price;
this.date = date;
   }
   public int getOrderID() {
 return orderID;
   }
   public void setOrderID(int orderID) {
this.orderID = orderID;
   }
   public double getPrice() {
       return price;
   }
   public void setPrice(double price) {
       this.price = price;
   }
   public String getDate() {
 return date;
   }

   public void setDate(String date) {
this.date = date;
   }
```

```
}
package restaurantsystem.model;
/**
 *
 * @author Aarya
 */
public class OrderLine {
   private int orderID;
private String name;
private int quantity;
private double price;
   public OrderLine(int orderID, String name, int quantity, double price) {
this.orderID = orderID;
  this.name = name;
 this.quantity = quantity;
this.price = price;
   }
   public int getOrderID() {
return orderID;
   }
   public void setOrderID(int orderID) {
 this.orderID = orderID;
   }
   public String getName() {
return name;
   }
   public void setName(String name) {
this.name = name;
   }
   public int getQuantity() {
       return quantity;
   }
   public void setQuantity(int quantity) {
  this.quantity = quantity;
```

```
    }
    public double getPrice() {
        return price;
    }
    public void setPrice(double price) {
        this.price = price;
    }
}
```

Assignment – 17:

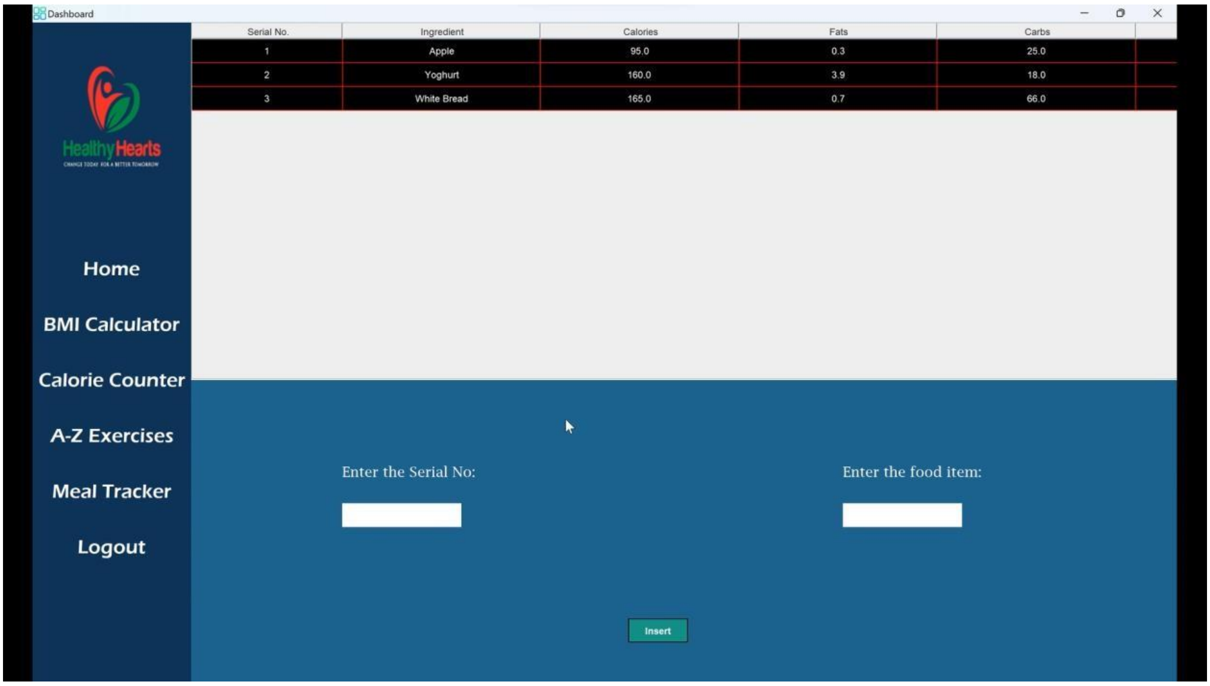
Writeup-2:

Healthy Hearts

Aim: Database connectivity and

backend analysis of Calorie counter.

Output:



Code:

```
String ingredient_name = tf.getText();  String
        serial_no_new = tfsr.getText();
        if(f.getSource()==b0) {
            double totalCalories = 0;
            double calorieCount = 0;
            double fatCount = 0; double
            carbCount = 0; double
            proteinCount = 0; Connection
            connection; try
```

```
connection = DriverManager.getConnection("jdbc:mysql://localhost:3306/diet", "root",
"Shawn@2004");
                    String sql = "SELECT calories_per_100g, fats_per_100g, carbs_per_100g,
 protein_per_100g  FROM FoodCalories WHERE ingredient_name = ?";
                    PreparedStatement preparedStatement = connection.prepareStatement(sql);
                    preparedStatement.setString(1,  ingredient_name);


                    ResultSet result = preparedStatement.executeQuery(); if
                    (result.next()) {
                        calorieCount = result.getDouble("calories_per_100g"); fatCount =
                        result.getDouble("fats_per_100g"); carbCount =
                        result.getDouble("carbs_per_100g"); proteinCount =
                        result.getDouble("protein_per_100g"); totalCalories +=
                        calorieCount;
                    } else {
                        JOptionPane.showMessageDialog(null,"Food Item not
 found","Error",JOptionPane.ERROR_MESSAGE);
                    }
                }
                catch (Exception ex) {
                    ex.printStackTrace();
                }
                Object[] newRow2 =
 {serial_no_new,ingredient_name,calorieCount,fatCount,carbCount,proteinCou nt};
                model2.addRow(newRow2);
```

```
            DefaultTableCellRenderer centerRenderer = new DefaultTableCellRenderer();
            centerRenderer.setHorizontalAlignment(SwingConstants.CENTER);

            for (int i = 0; i < table2.getColumnCount(); i++) {
table2.getColumnModel().getColumn(i).setCellRenderer(centerRenderer);
            }
        }

        tf.setText(null);
        tfsr.setText(null);
```